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SUPPLEMENTARY MATERIAL

Critical Overflow in Weakly Coupled Two-Choice Allocation

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This supplement documents the evaluation and finite-size comparisons in the main article. The main article gives the three main proofs; this supplement supplies the auxiliary estimates used there. The numerical bounds concern the preserved inputs and their stated arithmetic and probability contracts.

SM1 The centered Gini inequality

For a finite zero-mean real law Y,

$$\mathbb{E}|\sinh Y - \sinh Y'| \geq \mathbb{E}|\sinh Y| \geq \mathbb{E}\cosh Y - 1.$$

The second inequality holds pointwise. For the first, decompose a zero-mean law into zero-mean two-point laws across zero, plus any atom at zero, by transporting the positive first-moment mass against the negative first-moment mass. For a component on $a, -b$, $a, b > 0$, the difference between the two sides is

$$\frac{(a-b)(b\sinh a - a\sinh b)}{(a+b)^2} \geq 0,$$

since $\sinh(t)/t$ increases on $(0,\infty)$. The Gini functional of a real law is concave, as seen from $2\int F(t)(1-F(t))\,\mathrm{d}t$, whereas the absolute first moment is linear. Mixing the components proves the result. Truncation and mean correction extend it when the stated exponential moments are finite. Rank summation by parts then applies this inequality to suffix deficit $\delta t(1-t)$, giving the potential drift used in Sections 3 and 4.

SM2 Auxiliary process estimates

These estimates support the main proofs. The label-forgetting coupling, midpoint contraction, deterministic restarts, and joint-limit argument remain in the main article.

SM2.1 Dispersion bound

We prove main-article Lemma 3.1 with $\theta = 1/200$, $K_0 = 16/3$, and $I = (1-\varepsilon)/2$.

Proof. Order each community by increasing load, placing left buckets before right buckets at equal load. Use a fixed label order within each side. This order agrees with the left tie rule. For the heaviest k buckets in this order put $t = k/q$. If their two side counts are a, b, their internal selection probability is at most $ab/m^2 \leq t^2$. Cross mass entering this suffix is $x \leq \varepsilon t$, while cross mass entering the community is $y \geq x$. Conditional on a community update, its suffix probability is at most

$$\frac{It^2 + \varepsilon t}{I + \varepsilon t} \leq t - \delta t(1-t), \qquad \delta = \frac{I-\varepsilon}{I+\varepsilon} = \frac{1-3\varepsilon}{1+\varepsilon} \geq \frac{1}{5}.$$

The global inclusion probability of a vertex is $1/q$, so its conditional update probability is at most $8/(3q)$. Rank summation by parts, for sorted $g_i = \sinh(\theta\xi_i)$, gives

$$\sum_i (p_i - q^{-1})g_i \leq -\frac{\delta}{2}\mathbb{E}|\sinh Y - \sinh Y'| \leq -\frac{\delta\Gamma_s}{2q} + \frac{\delta}{2}.$$

Here Y is uniform on the centered $\theta\xi_i$ and Y' is independent. The zero-mean inequality $\mathbb{E}|\sinh Y - \sinh Y'| \geq \mathbb{E}|\sinh Y| \geq \mathbb{E}\cosh Y - 1$ is proved in Section SM1. For the centered increment $u_i = \mathbf{1}_{i \text{ selected}} - 1/q$, its second moment is less than $3/q$. Taylor's formula bounds the total cosh remainder by $4\theta^2\Gamma_s/q$. Since $\theta \leq \delta/16$,

$$\mathbb{E}[\Delta\Gamma_s \mid C_s \text{ updates}] \leq -\frac{\theta\delta}{4q}\Gamma_s + \frac{\theta\delta}{2}.$$

Each community updates with probability at least $3/8$. The total drift is therefore at most

$$-\frac{3\theta\delta}{32q}\Gamma + \frac{\theta\delta}{2}.$$

The recurrence from $\Gamma_0 = 2q$ proves (3.1). Only the cross marginals, rather than their independence, were used. □

SM2.2 Fixed-window empirical estimate

Use the translated initial state and constants in main-article section 5. The mean clock is $u = k/q$, and the fluid starts from the exact empirical profile. We prove the uniform estimate (5.1) for fixed B, R.

A valid explicit bound in (5.1) is

$$\eta_q(B,R,K_1) = e^{4B}\left[8(2K_1+1)\sqrt{B/q} + 4BC_{R,B}e^{-\theta(K_1-1)} + 8B/q\right],$$
$$C_{R,B} = 2e^{\theta}R\exp\{2B(e^{\theta}-1)\}.$$

Proof. Each event increments exactly one tail coordinate by $2/q$. Its conditional mean increment is exactly $F(X_k^q)/q$, where F is the field in (2.1). Subtract these conditional means to obtain a tail-valued martingale at the grid times. Each coordinate has expected quadratic variation at most $4B/q$. Doob's inequality bounds its expected maximum by $4\sqrt{B/q}$. There are $2(2K_1+1)$ retained coordinates, indexed by $|k| \le K_1$.

A bucket has selection probability at most $2/q$ at each internal event. Therefore the expected sum of its translated, uncentered cosh moments grows by at most the factor $1 + 2(e^{\theta}-1)/q$ per event. The initial sum of the two side moments is at most $e^{\theta}R$. The probability that the selected tail increment lies outside the retained coordinates is at most $C_{R,B}e^{-\theta(K_1-1)}$. This follows from the exponential moment bound and the selection-probability bound. Summing actual outside increments and their nonnegative compensators gives expected outside martingale norm at most $4BC_{R,B}e^{-\theta(K_1-1)}$. Initial tails add no martingale error, since the fluid starts from the exact empirical profile.

The field is 4-Lipschitz in tail ℓ_1. A single interpolated empirical step has length at most $2/q$. Replacing the stepwise field integral by the interpolated one costs at most $8B/q$. Gronwall gives (5.1). □

SM2.3 Moments for profile tracking

The following calculation gives main-article (5.2).

For the real product process, let W_n be the sum of its four side Wasserstein distances to the corresponding orbit laws. Each community uses its actual mean $M_s(n)/q$. Let Ψ_n be the sum of the four empirical cosh moments centered at these means. Lemma 3.1 implies

$$\sup_n \mathbb{E}\Psi_n \le C_E = 32/3.$$

If J_n is the sum of the four empirical mean absolute centered deviations, Jensen gives

$$e^{\theta J_n/4} \le \Psi_n/2, \qquad \mathbb{E}J_n^2 \le 16C_E/\theta^2.$$

The second inequality uses $x^2 \le 2e^x$ for $x \ge 0$. The four orbit absolute deviations sum to at most $8/\theta$, since $V_\theta(\rho^u) \le 4$. Thus

$$\sup_n \mathbb{E}W_n^2 \le C_*^2 = (32C_E + 128)/\theta^2. \tag{SM2.1}$$

SM2.4 Label forgetting

This proves main-article Lemma 3.2, using its constants and notation.

Proof. Write P for the fixed transition kernel and $\mu = \mu_{n-L}$. Split

$$\mathrm{TV}(P^L\mu, \tau P^L\mu) \le \mathrm{TV}(P^L\mu, P^L\tau\mu) + \mathrm{TV}(P^L\tau\mu, \tau P^L\mu).$$

These two terms need different couplings. For the first, start A from μ and A' from τA. Map internal candidates by τ until the two tagged loads in A meet. This is legal uniform internal sampling and maintains $A' = \tau A$. Use the same actual cross edge; the relation can fail only when a cross touches a tag, with window probability at most $2\varepsilon\ell$. Once the tag loads meet the configurations agree, so common candidates keep them equal.

Before meeting or a bad cross, the higher tag has no larger internal update rate: a left tag at h has rate $(1-\varepsilon)y_h/q$, a right tag has rate $(1-\varepsilon)x_{h+1}/q$. The embedded positive gap is consequently dominated by a fair nearest-neighbor walk absorbed at zero. Its survival for J jumps from gap d is at most $\min(1, 4d/\sqrt{J+1})$, by reflection and the central binomial bound. Fresh direction coins give this domination in the adapted environment; we never condition those coins on a future update count.

Each external uniform tag has expected centered cosh at most $2K_0$, hence expected initial gap at most $4K_0/\theta$. The community receives at least $L/4$ internal arrivals except with probability $e^{-L/48}$, by the binomial lower tail since $I \ge 3/8$. At the two window endpoints, all four tag deviations are at most $\ell/16$ except with probability $16K_0e^{-\theta\ell/16}$. Off these events, each tag gains at least $\ell/8$, since the community mean gains at least $\ell/4$. Put $K = \lceil \ell/4 \rceil$. If the tags have not met and no cross touches them, at least K embedded internal jumps have occurred. The nonmeeting event is therefore contained in the union of the stated bad events and survival of the dominating fair walk for the fixed number K of jumps. No conditioning on the final random jump count is used. Reflection and $\mathbb{E}d \le 4K_0/\theta$ bound this last probability by $32K_0/(\theta\sqrt{\ell})$. This controls the first TV term with an additional $2\varepsilon\ell$.

For the second term compare the fixed kernel with its τ-conjugate, from the same law $\tau\mu$. Their internal laws agree; the two swapped cross columns have total conditional mass $2/m$. Coupling all other mass gives another window cost $2\varepsilon\ell$. The reasoning also holds for a weighted doubly stochastic matrix. Adding the costs proves (3.2). The separate conjugate-kernel term is necessary: forgetting a swapped initial state alone does not imply invariance of the final law. □

SM3 Auxiliary numerical estimates

SM3.1 Analytic tails for the true orbit

The posterior certificate requires bounds on the infinite true orbit, not only on the saved finite path. Here are the bounds actually used. Let $D_{LR}(u) = \mathbb{E}|L_u - R_u|$ with independent draws. Exact differentiation of the centered second-moment sum gives

$$\frac{\mathrm{d}}{\mathrm{d}u}\{\mathbb{E}(L_u - u)^2 + \mathbb{E}(R_u - u)^2\} = 2 - 2D_{LR}(u).$$

It follows by periodicity that $\int_0^1 D_{LR}(u)\,\mathrm{d}u = 1$. This function is 2-Lipschitz by the monotone marginal coupling, so its maximum is at most $3/2$ on the unit circle. Interlacing gives $0 \le \mathbb{E}L_u - \mathbb{E}R_u \le 1$. Jensen and positive/negative parts, at mean zero, then imply

$$\mathbb{E}(L_0)_+ \le 5/4, \quad \mathbb{E}(R_0)_+ \le 3/4, \quad \mathbb{E}(-L_0)_+ \le 3/4, \quad \mathbb{E}(-R_0)_+ \le 5/4.$$

For example $\mathbb{E}(L-R)_+ = (D_{LR} + \mathbb{E}L - \mathbb{E}R)/2$, while Jensen bounds the marginal positive part by that expression after accounting for the other marginal's sign of mean. The opposite-sign cases follow identically.

Integrating the front equation and matching the limit at $+\infty$ yields

$$F(r) = \int_{r-1}^{r} F(s-1)F(s)\,\mathrm{d}s, \qquad F(r) \le F(r-1)F(r-2). \tag{SM3.1}$$

The first-moment bounds give $F(2k) \le \min(1, 5/(4k))$ and $F(2k+1) \le \min(1, 3/(4k))$ for $k \ge 1$. Monotonicity and (SM3.1) give a rational superlinear right-tail recursion. Once $F(n) \le \eta < 1$, the uncomputed remainder obeys $F(r) \le \eta F(r-2)$ for $r \ge n+2$. Upward rounding and this geometric remainder account for the infinite tail even when finite arithmetic plateaus.

For the left anchor set $z_j = (x_j(0) + y_j(0))/2$ and $A_j = \int_0^1 x_j(u)y_j(u)\,\mathrm{d}u$. The pooled equation and the unit-period shift give $A_j = z_j$. Stochastic increase gives

$$z_j \le x_{j-1}(0)y_{j-1}(0) \le z_{j-1}^2.$$

If $z_{-2} \le s$, mean zero would require $3 \le \sum_{n\ge 0} s^{2^n}$. At $s = 89/100$ the latter sum is strictly below three: sum through $n = 5$ exactly and bound the rest by $s^{64}/(1 - s^{64})$. Thus $G(-4) = 1 - F(-4) < 11/100$.

Write $a = 11/100$, $c = 1 - a/2 = 189/200$ and $\gamma = 18/25$. From (SM3.1), for $r \le -4$,

$$G(r) = \int_{r-1}^{r} [G(s) + (1 - G(s))G(s-1)]\,\mathrm{d}s \ge c\int_{r-2}^{r} G(s)\,\mathrm{d}s.$$

The kernel mass $c\int_0^2 e^{-\gamma s}\,\mathrm{d}s$ is at least one, as verified by a rational Taylor lower bound $e^{36/25} > 21/5$. Therefore $e^{-\gamma r}G(r)$ is superharmonic for the backward walk with normalized density proportional to $ce^{-\gamma s}$ on $[0,2]$. Starting at -4, stop on entering $[x, x+2]$. The stopped function is bounded on the traversed compact interval and the stopping time is finite, so the supermartingale inequality gives

$$G(x) \le \frac{11}{100}\exp\{(18/25)(x+6)\}, \qquad x \le -6.$$

The walk is only an analytic proof device; it is never simulated. For $K_2 \ge 3$ its integrated contribution is

$$t^-_{\rho^0}(K_2) \le \frac{11}{100}\frac{b^{2K_2-4} + b^{2K_2-5}}{1 - b^2}, \qquad b = e^{-18/25}.$$

The rational right recursion and this geometric left bound are used in (SM4.1). An independent coarser bound with exponent $3/5$ was retained as a fallback. All scalar comparisons precede the matrix stages.

SM3.2 Validated lower matrices and bootstrap

At saved endpoints, tail values are enclosed by the exact binary rationals plus or minus the strict full-field trajectory error. Across a cell, true tails are monotone in mean time. An atom bound over a cell of width h is $\max\{0, \widehat{a}_l - e_l\}\,\underline{e^{-2h}}$, where e_l bounds the left-endpoint atom error and the exponential is bounded below. Over candidate cycle k, nonexpansion and the period residual add at most kr to the discrepancy from the integer-translated first cycle.

The orbit is first enclosed analytically using the preceding tails. After a completed stage gives $E_{\mathrm{old}} \geq d_1(v, \rho^0)$, nonexpansion also encloses every orbit phase by first-cycle nodes with error $e_s + E_{\mathrm{old}}$, independently of cycle index. Intersect these valid bounds. A stage never assumes its own proposed result.

The midpoint-chain lower off-diagonals are the sums of the compared-law lower tails and atoms, because the rate twice a midpoint is that sum. Negative diagonals use upper exit bounds, rather than replacing every exit by two. After each period shift the finite frame by one and kill any path leaving it. For $A = -2I + C$, downward arithmetic computes

$$e^{hA} \;\geq\; \underline{e^{-2h}} \sum_{j=0}^{4} \frac{h^j C^j}{j!}$$

entrywise, with a nonnegative finite right side. Propagating all core columns and summing row minima supplies the stated common mass. The backward verifier uses its own validated bootstrap history. Both histories and their initial target failures are retained; the final readout uses their larger error bound.

SM4 Numerical certificate details

Let v be an exact rational probability profile of mean zero, obtained by normalizing the saved candidate and correcting its mean with an integer-shift mixture. Binary64 input values are treated as exact rationals, not as exact solutions. Its strict one-period return error is $r < 4.601\,10^{-9}$. The full-cell trajectory error is e_s, including the mean correction; $e = \sup_s e_s < 4.585\,10^{-9}$. Bounds come from the full polynomial residual, including the top outflow. For a Bernstein residual of degree six, integration of its absolute value is bounded by the cell width times the sum of absolute coefficients divided by seven. Tail nonexpansion then controls the error against the true flow of v.

Proposition SM4.1 (A posteriori profile estimate)**.** Fix a core $|k| \leq K_2$ on both channels. Suppose the exact midpoint difference kernel between the candidate flow and the orbit, over B mean periods with the co-moving shifts, has common mass at least $\beta > 0$ from every core starting state. For

$$t_p(K_2) = \sum_{\text{sides}} \{\mathbb{E}_p(X - K_2)_+ + \mathbb{E}_p(-X - K_2 - 1)_+\},$$

one has

$$d_1(v, \rho^0) \leq Br/\beta + 2t_v(K_2) + 2t_{\rho^0}(K_2). \tag{SM4.1}$$

Proof. The signed tail difference has total mass zero. Its outside-core absolute mass is at most $t_v + t_{\rho^0}$. On the core the common kernel mass cancels at least β times the smaller of its positive and negative masses, reducing the ℓ_1 norm by at least $\beta[d_1 - 2(t_v + t_{\rho^0})]$. Iterating the one-period return bound and nonexpansion gives $d_1(v, T^B v) \leq Br$. Triangle inequality between $v, T^B v, \rho^0 = T^B \rho^0$ now yields (SM4.1). In particular the tail term is not divided by β. □

The finite lower kernels retain only paths inside their declared moving frame, so deleting escaping paths preserves a lower bound. Validated exit upper bounds avoid unnecessary killing at high starting levels. Each lower Metzler matrix has form $-2I + C$, $C \geq 0$. Downward dyadic arithmetic, the nonnegative exponential terms through degree four, and a rational lower bound on e^{-2h} propagate all core columns. At each final row take the minimum over all core columns and sum these minima to obtain β. An independent backward implementation propagates all terminal indicators with 64-bit directed arithmetic; the forward computation uses 80 bits. Each bootstrap stage uses only an already completed earlier bound. Supplement section SM3.1 gives the analytic infinite-tail bounds required by (SM4.1).

The larger final forward/backward bound is the exact rational

$$E = \frac{1365032393294082761}{151115727451828646838272} < 9.034\,10^{-6}.$$

Nonexpansion and integer translation give uniform profile error $D_* = E + e < 9.038\,10^{-6}$ at every real mean. Negative means use translation, not backward integration.

For feedback, the origin coefficient and overflow, exact rational sums over 257 saved phase nodes (256 intervals) yield errors

$$\text{feedback: } 2D_* + \frac{9472}{180 \cdot 256^4}, \qquad \text{origin coefficient: } 8D_* + \frac{76288}{180 \cdot 256^4}, \qquad \text{overflow: } D_*.$$

The profile constants follow by coupling the two independent comparisons, a three-draw origin event, and positive part. For quadrature, the quadratic field has $\|F\|_1 = 2$, $\|DF\| \le 4$, $\left\|D^2F\right\| \le 8$ and higher field derivatives zero. Its trajectory derivative norms through order four are bounded by $2, 8, 64, 640$. Multinomial differentiation gives fourth derivative bounds 9472 for the feedback integrand and 76288 for the origin integrand. Composite Simpson quadrature has error $M_4/(180N^4)$; its positive weights do not amplify grid profile errors.

Table SM1 Preserved certificate stages. The first two stages miss the original 0.001 target. The fourth reaches the stricter 0.0001 stopping target. All distance bounds are rounded upwards.

Core radius	Periods	Frame radius	Subdivisions	Common mass	W_1 bound
4	8	8	64	0.0006913	0.050068
6	16	10	64	0.0651929	0.002797
8	24	12	128	0.5510914	0.000157
10	32	14	128	0.5811574	0.000009034

Table SM2 Deterministic readouts of the limiting profile. Intervals include profile and quadrature errors. They do not bound finite-allocation error.

Readout	Certified interval
$-\overline{g}'(0)$	$[0.71997987,\ 0.72012468]$
$\overline{g}(1)$	$[-0.62242623,\ -0.62239004]$
$\overline{g}(2)$	$[-0.90371675,\ -0.90368056]$
$S_0(0)$	$[0.63127654,\ 0.63129462]$
$S_0(1)$	$[0.75422265,\ 0.75424073]$

SM5 Statistical protocols and interpretation

Coverage assumes uniform independent underlying words within each stream and across replicate labels. Fresh microscopic streams must also be independent of prior history. Counter-based SplitMix words and disjoint seeds provide reproducibility, not a proof of those premises. All sampling statements use this ideal word model.

The original and prospective families below are separate. Reusing a reference retains the reference event's original error cost. Cross-cell independence is unnecessary for the union bounds. The prospective statement is unconditional over the old reference and fresh data, rather than conditional on the observed reference.

SM5.1 Original allocation design

The model is exactly Section 2, with identity matchings in both directions or product crosses, $q \in \{64, 128, 256, 512\}$, $\lambda \in \{0, 1, 2\}$ and $\varepsilon = \lambda/q$. Capacity is the integer $c = q/2$, so $\kappa = 1$ and $n_* = q^2$. The respective replicate counts are 128, 128, 256, 1024. Every paired path runs through $q^2 + 2q$ arrivals. There are 4608 fixed/product pairs, or 9216 individual outcomes. Every planned label was completed; none was replaced or excluded.

The 1024 seed labels are the first eight SHA256 bytes of the fixed string prefix and replicate number specified in the protocol, interpreted as a big-endian 64-bit word. Smaller scales use prefixes of this list. A counter-based SplitMix64 driver supplies two words per arrival; disjoint bit fields determine event type, direction and endpoints. All probabilities are exact power-of-two subdivisions at these parameter values. The same driver words are used within a fixed/product pair and across cells sharing a replicate label. Thus cells and checkpoints are dependent. The seed label, not a checkpoint, is the inferential unit. The confidence guarantees below are conditional on the declared ideal independent-stream reference. Reproducible pseudorandom words do not establish mathematical independence of machine streams.

Compilation and deterministic one-step tests preceded the one authorized stochastic batch. The model checks mass and capacity identities, endpoint counting, and left-tie handling. All 9216 full labelled critical configurations were subsequently re-read and independently checked against their mass, imbalance and overflow summaries. A second read-only audit reconstructed all 193536 saved checkpoint masses, overflows, product feedback counts and fixed joint-table marginals, then all 9216 outcome metrics; its largest metric roundoff difference was below $2.3\,10^{-16}$. At $\lambda = 0$ all 32256 paired recorded checkpoints agree, as required. All checkpoint side histograms, fixed joint tables, driver hashes and logs are retained. The single-thread model batch took about 22.35 seconds of wall time within its fixed 360-second CPU and 420-second wall budgets. This duration is operational evidence, not an algorithm performance benchmark; no competing timed runs were active.

For phase-aware feedback there are 16 prescribed checkpoints

$$n_j = q^2/2 + 2q\left\lfloor \frac{(q/4-1)j}{15} \right\rfloor + \frac{2qj}{16}, \quad j = 0, \dots, 15.$$

The common phase is exactly $j/16$. The metric is the trajectory RMS of the exact conditional feedback minus $g(h_n, \Delta_n/q)$, not minus $\overline{g}$. For overflow the metric is the maximum absolute error of $Z_c/q - S_h(\Delta/q)$ over the five fixed h values $-1, -1/2, 0, 1/2, 1$. The numerical lookup uses 257 old phase nodes with linear tail interpolation. Its uniform allowances are 0.00004861 for feedback and 0.00002440 for overflow, including $E+e$, interpolation and arithmetic. The interpolation addition in d_1 is $1/65536$, since the tail second-derivative norm is at most eight. Numerical allowances are used on both sides of each classification.

■ SM5.2 Original failure-rate and cross-kernel intervals

Formal feedback/overflow and cross-kernel CDF tolerance decisions apply only at $q = 512$; $q = 64, 128, 256$ are descriptive. Four feedback cells ($\lambda = 1, 2$ and both kernels) use RMS margin 0.10. Six overflow cells (all three λ and both kernels) use maximum-error margin 0.05. For each cell the target is a population probability of exceeding the margin at most 0.10. A numerically certain failure exceeds the margin even after subtracting its allowance; a possible failure exceeds it after adding the allowance. Exact binomial tail inversion on a conservative 0.001 grid bounds the failure probability below using certain failures and above using possible failures. Each side uses error probability $1/800$, giving simultaneous coverage at least 0.975 over the ten cells by union bound. Entirely interior intervals support the target, lower bounds above it detect mismatch, and all other cases are inconclusive.

Table SM3 All primary $q = 512$ cells, with 1024 replicates per cell. All ten support their frozen population target. Overflow failures are present and are retained. Intervals include the deterministic numerical allowance through certain/possible counts.

Metric	λ, kernel	Certain	Possible	Probability interval
Feedback RMS	1, fixed	0	0	$[0, 0.007]$
	1, product	0	0	$[0, 0.007]$
	2, fixed	0	0	$[0, 0.007]$
	2, product	0	0	$[0, 0.007]$
Overflow maximum	0, fixed	37	37	$[0.020, 0.058]$
	0, product	37	37	$[0.020, 0.058]$
	1, fixed	38	38	$[0.021, 0.059]$
	1, product	41	42	$[0.023, 0.064]$
	2, fixed	49	50	$[0.030, 0.073]$
	2, product	45	47	$[0.026, 0.070]$

Cross-kernel comparison uses endpoint CDFs of Δ_{q^2}/q and $Z_c(q^2)/q$ at $\lambda = 1, 2$. It does not use raw common-random-number path distances: the driver coupling is not the maximal-load-pair rank coupling of Section 3. The frozen practical CDF-distance margin is 0.20. DKW–Massart bounds [1] of width 0.057 for each of the eight empirical CDFs yield simultaneous pair bands 0.114, with family error at most 0.025 because

$$16 \exp[-2 \cdot 1024 \cdot (0.057)^2] < 0.025.$$

The scalar inequality was checked with exact rational Taylor bounds. Independence between the two kernels is unnecessary for the union bound. Observed KS distances, in order $\lambda = 1$ imbalance/overflow and $\lambda = 2$ imbalance/overflow, are $9/1024$, $17/1024$, $13/1024$ and $1/64$. The true CDF-distance upper bounds are respectively 0.122790, 0.130602, 0.126696 and 0.129625 after upward rounding. All are below 0.20, supporting this specified finite-size equivalence criterion. The lower bounds are zero; no positive separation is established. Adding the primary and CDF family errors bounds the total simultaneous error by 0.05 under the reference model.

■ SM5.3 Prospective design

The prospective protocol fixes $q = 512$, $\kappa = 1$, $\lambda = 1, 2$, matching/product kernels and 4096 shared replicate labels. Each path stops at q^2 arrivals. The source freeze is dated 17 October 2026, 23:42:34 UTC. The input freeze precedes the first trajectory, and the data freeze precedes inference.

For label i, form the string

```
paper_micro_prospective_v1|replicate=%05d
```

with i formatted as five digits. Interpret its first eight SHA256 bytes as a big-endian 64-bit seed. All 4096 values are unique and disjoint from the original and reference seed lists. The frozen protocol permits no stochastic pilot, replacement, extension or threshold change. The artifact retains the complete batch and its checks.

■ SM5.4 Prospective intervals and coverage allocation

The fixed-time nonnegative-product argument is a simple instance of betting-based mean inference [2]. For means, clip $V_i = \min(Z_i/q, 3)$ only for inference and retain all unclipped observations. Lower candidates $m \in [0, 1]$ use $\prod_i [1 + V_i - m]$; upper candidates $m \in [1/2, 3]$ use $\prod_i [1 + (2/5)(m - V_i)]$. Reject only at capital at least 1000, and invert on the fixed 10^{-5} grid outwards using exact integer products. Each endpoint costs at most 0.001 by the fixed-time nonnegative-product bound; no normal approximation is used.

Changing one arrival changes final labelled-load ℓ_1 distance by at most two and overflow by at most one. At q^2 arrivals, bounded differences thus gives $\mathbb{P}(X - \mathbb{E}X > t) \le e^{-2t^2}$ for $X = Z/q$. The fixed coarse bound $U = \overline{X} + 0.030$ costs at most $e^{-2R(0.030)^2} < 0.001$ per cell. On $\mathbb{E}X \le U < 3$, add the upward-evaluated clipping-bias bound

$$\mathbb{E}(X - 3)_+ \le \frac{1}{4(3 - U)} e^{-2(3-U)^2}.$$

All four mean-width gates ≤ 0.04 passed. No sample exceeded three, but the population tail allowance was still included.

The existing reference's four fixed mean endpoints each cost at most 0.001. Its clipped mean intervals are $[0.69875, 0.70333]$ at $\lambda = 1$ and $[0.67391, 0.67781]$ at $\lambda = 2$. The audited total approximation and truncation allowances 0.0006 and 0.0010 give true SDE mean intervals $[0.69815, 0.70393]$ and $[0.67291, 0.67881]$. Both satisfy the unchanged width gate 0.010. Subtraction of these intervals from the fresh unconditional micro intervals gives Table 1. Eight fresh betting endpoints, four fresh coarse bounds and four existing reference endpoints cost at most 0.016.

Fresh micro CDFs use DKW width $a = 0.032$ with $R = 4096$; the two existing reference CDF events use $b = 0.012$ with $N_{\text{ref}} = 32768$. The audited reference-to-SDE coupling supplies horizontal radius $r_1 = 0.010$, $r_2 = 0.020$ and vertical allowance $\tau = 0.005$. With A the micro empirical law, B the computed reference empirical law, $B_\pm = B \pm r_\lambda$, $d(A, B) = \sup_x (F_A - F_B)$ and $\delta = a + b + \tau = 0.049$, the distance bounds are

$$\begin{aligned} L &= \max\{0, \max[d(A, B_-), d(B_+, A)] - \delta\}, \\ U &= \min\{1, \max[d(A, B_+), d(B_-, A)] + \delta\}. \end{aligned}$$

Every jump, including ties, is included; stored long-double hexadecimal values are parsed as exact rationals. The unchanged reference CDF-envelope width gate 0.10 passed for both λ values. The simultaneous failure allocation for this family's eight comparisons is

$$\begin{aligned} \alpha_{\text{mean}} &\le 0.016, \\ \alpha_{\text{CDF}} &\le 8e^{-2\cdot 4096(0.032)^2} + 4e^{-2\cdot 32768(0.012)^2}, \\ \alpha_{\text{joint}} &< 0.018138367441 < 0.05. \end{aligned}$$

The scalar exponential bounds were certified before sampling with rational Taylor bounds. This is at least 95% simultaneous coverage for these eight new comparisons under the contracts below. It is a separate family from the original finite-batch claims, not a combined confidence statement for every result in the manuscript. Between-cell independence is not needed for the union bounds.

■ SM5.5 Earlier post-hoc diagnostic kept separate

Before the prospective batch, the old $R = 1024$ micro endpoints had already been observed when an SDE follow-up recipe was designed. Its micro mean intervals are fixed-recipe nominal records; coverage adjusted for that actual post-hoc selection was not established. The original unqualified whole-family 95% statement for that diagnostic was withdrawn. The separately documented CDF-only anchor does not repair the micro mean selection gap. See the preserved corrected diagnostic status.

■ SM6 SDE reference error budget

Write $b = \lambda \overline{g}$, $N = 16384$ and $h = 1/N$. The front representation gives $|b| \le \lambda$, $-2\lambda \le b' \le 0$ and $|b''| \le 24\lambda$. Thus $x + hb(x)$ is nondecreasing and 1-Lipschitz. Let $L = 2\lambda$ and $C_0 = 2\lambda^2 + 12\lambda$.

The local exact-versus-Euler remainder has the Itô representation

$$R_k = \int_0^h (h - u) b'(X_{kh+u}) \, \mathrm{d}B_u + \int_0^h (h - u)(bb' + b''/2)(X_{kh+u}) \, \mathrm{d}u.$$

Its conditional mean is bounded by $a = C_0 h^2/2$. Its centered conditional L_2 norm is at most $s = Lh^{3/2}/\sqrt{3} + C_0 h^2/2$. Contraction yields the second-moment recurrence

$$v_{k+1} \le v_k + 2a\sqrt{v_k} + a^2 + s^2.$$

Induction gives $\sqrt{v_N} \le \sqrt{N}s + Na$. Hence the Euler bound has an explicit constant and no exponential amplification.

The global feedback table uses Simpson quadrature over 256 phase intervals and linear interpolation at spacing $1/512$. Its error is at most

$$0.00004861 + \frac{9472}{180 \cdot 256^4} + \frac{3}{512^2} < 0.000061.$$

For S_0, linear interpolation adds $1/(4 \cdot 512^2)$ to 0.00002440, giving error below 0.000026. Certified analytic tails cover constant feedback extension beyond eight and the overflow extension $|d|/2$ beyond 32. All approximate feedback slopes lie in $[-2, 0]$, and all approximate overflow slopes lie in $[0, 0.5001]$.

For midpoint uniforms of cell width $\delta = 2^{-52}$, an explicit Box–Muller coupling gives

$$\mathbb{E}|Z - Z_{\mathrm{mid}}|^2 \le 100\delta(1 + \log(1/\delta)) < 10^{-12}.$$

The implementation's series and rounding bounds are stated below. Minkowski across N scaled increments gives accumulated driver error at most $128(10^{-6} + 10^{-9}) < 0.00013$. This bound does not assume cancellation of coordinate errors.

For clipping at three, the ideal-Gaussian Euler endpoint is a 1-Lipschitz function of its Gaussian increments. Oddness and contraction give its second moment at most one. The approximate readout has value below 0.632 at zero and Lipschitz constant 0.5001. Its mean is below 1.133. Gaussian concentration [3] gives clipping bias below

$$\frac{0.5001^2}{3 - 1.133} \exp\left[-\frac{(3 - 1.133)^2}{2(0.5001)^2}\right] < 0.00014.$$

The reference error allowances include the globally certified feedback/readout tables, Euler discretization, the Gaussian driver, state/readout arithmetic and truncation. The coefficient-table allowances are 0.000061 for $\overline{g}$ and 0.000026 for S_0, with checked readout Lipschitz bound 0.5001. The Gaussian-driver accumulated L_2 allowance is 0.00013. The authoritative arithmetic supplement bounds normal-coordinate arithmetic error below 10^{-12}, supporting the frozen 10^{-9} coordinate allowance; state rounding plus readout rounding divided by 0.5001 is below $4\,10^{-12}$, within the single shared 10^{-6} allowance. These statements are conditional on binary round-to-nearest arithmetic with at least 64 significand bits, normal finite operation ranges and exact radix-two exponent reduction. No historical dynamic floating-point register snapshot was saved for the old reference; a later environment probe cannot prove those historical registers. The audited version 2 arithmetic supplement is included with the source, together with the reference protocol and probability contract.

For clarity, the reference mean budget uses two different readout constants. The exact S_0 is $1/2$-Lipschitz, whereas the stored interpolator is 0.5001-Lipschitz. Let e_λ denote the certified Euler L_2 bound. With $N = 16384$, $L = 2\lambda$ and $C_0 = 2\lambda^2 + 12\lambda$, the monotone Euler-map estimate gives

$$e_\lambda = \frac{L/(17/10) + C_0/2}{N} + \frac{C_0}{2N\sqrt{N}},$$

using $\sqrt{3} > 17/10$ and $\sqrt{N} = 128$. For $\delta_g = 0.000061$ and $\delta_S = 0.000026$, the mean error is bounded by

$$\delta_S + \tfrac{1}{2}(e_\lambda + \lambda\delta_g) + 0.5001(0.00013 + 0.000001) + 0.00014.$$

It is less than 0.000514 at $\lambda = 1$ and 0.000857 at $\lambda = 2$. These bounds fit the frozen allowances 0.0006 and 0.0010. The CDF coupling follows from

$$(e_\lambda + 0.00013)^2 \le 0.005(r_\lambda - \lambda\delta_g - 0.000001)^2.$$

The first two terms in the mean bound compare the true SDE to the ideal-Gaussian Euler process. The remaining terms cover the computed driver, joint arithmetic error and Gaussian clipping tail. The arithmetic supplement supplies the local rounding bounds; no cancellation of driver errors is assumed.

■ SM7 Artifacts and verification scope

The reproducibility repository is `https://github.com/19991111/critical-overflow-reproducibility`. Selected frozen inputs are supplied in this repository. Its file-level provenance map records their original commits, archive members, and checksums. The source commits below identify the original runs.

The figures and comparisons use preserved experiments and numerical certificates. Their preparation adds no allocation trajectory, seed, or numerical integration. All original archives and earlier manuscript versions remain separate.

The original allocation batch contains 9216 outcomes and 193536 checkpoint observations. Its original source commit is

`dddf58f43c0831e1c24e3017fc11618938453478`.

The prospective batch contains 16384 endpoints from 4096 shared labels. Its artifact map identifies source, protocol, seeds, configurations, hashes and logs. The original source commit is

`09480b6e9afcd54574245433df21c1cfac1eede6`.

The SDE reference originates from

`067543228ccdc4efbce5b8df7a0849b03f5fc1ec`.

Its reused status and original confidence events are part of the comparison design.

The accompanying `supporting_records_fcs.zip` contains the authoritative arithmetic supplement, probability contract, reference protocol, frozen prospective plan, and corrected post-hoc status. The same records are included in the `evidence/` directory of the LaTeX source bundle. Its `RECORDS_INDEX.txt` maps these files to the supplement sections. File manifests identify the exact input bytes. Local audit receipts record the saved-output checks.

The completed numerical audit independently recomputes all 256 full-cell residual bounds, 24 fluid readout estimates, 20482 coefficient nodes and 65536 reference endpoint readouts. It exactly reduces the saved forward and backward terminal matrices. It does not replay matrix propagation. That stage relies on the inspected comparison proof and archived implementation.

The endpoint audit recomputes mass, imbalance and overflow from all 25600 saved allocation configurations. The statistical audit recomputes prospective betting endpoints and neighboring grids, clipping bounds, CDF suprema and family allocations. It also checks the original ten failure-rate intervals and four cross-kernel CDF comparisons. The earlier experiment artifact records a separate checkpoint reconstruction; the later audit does not repeat that full reconstruction.

Replicate labels are shared across cells, so 16384 prospective endpoints represent 4096 labels in four cells. At $\lambda = 0$, the original matching and product outcomes coincide under the shared driver. Both records are retained, but they do not provide independent replication. All reported summaries are computed within cells; the confidence statements use the union bounds in Section SM5.

The recorded freeze hashes and chronology are internally consistent. They are not an external timestamp attestation. The old reference has no saved dynamic floating-point register snapshot. Its arithmetic guarantee therefore remains conditional on the contract in Supplement section SM6. Pseudorandom reproducibility does not establish the ideal independence premises.

The LaTeX source bundle includes build instructions. The supporting records retain the numerical and statistical audit receipts. Internal mathematical review and computational checks are not machine formalization or external peer review. The experiments measure allocation overflow and the stated approximation errors.

SM8 Detailed contraction and attraction proof

We supply the analytic details behind main-article section 4. Fix $\theta = 1/200$. All constants in the positivity argument below may depend on the initial laws and the finitely many states being considered. No lower transition bound uniform over the lattice or all initial laws is assumed.

SM8.1 Flow, moments, and compact sets

The atom equations associated with the tail flow are

$$a_k' = 2a_{k-1}y_{k-1} - 2a_k y_k, \qquad b_k' = 2b_{k-1}x_k - 2b_k x_{k+1}. \tag{SM8.1}$$

These are birth equations with rates at most two. On the affine tail space of finite-first-moment laws, the tail field is globally 4-Lipschitz. One can construct the flow by Picard iteration for the corresponding two inhomogeneous birth laws. Bounded rates give nonexplosion and preserve first moments on bounded time intervals. Uniqueness follows from the Lipschitz bound. In particular, the atoms and tails are continuous in time, and the mean clock satisfies $\mu' = 1$.

Here is a direct verification of the moment estimate used for compactness. Let $\nu_k = (a_k + b_k)/2$ and $w_k = a_k y_k + b_k x_{k+1}$. Then $\sum w_k = 1$, $w_k \le 2\nu_k$, and $\sum_{j\ge k} w_j = x_k y_k \le [(x_k + y_k)/2]^2$. Thus w is stochastically below the minimum of two independent draws from ν. For $Y = \theta(X - \mu)$, $X \sim \nu$, and an independent copy Y', this gives

$$\sum_k (w_k - \nu_k)\sinh(\theta(k-\mu)) \le -\tfrac{1}{2}\mathbb{E}|\sinh Y - \sinh Y'| \le -\tfrac{1}{2}(\mathbb{E}\cosh Y - 1).$$

The last inequality is proved in Supplement section SM1. Taylor's formula gives $\cosh(z+\theta) - \cosh z - \theta\sinh z \le (e^\theta - 1 - \theta)\cosh z$. Differentiating the moving-center moment, with $V = 2\mathbb{E}\cosh Y$, therefore yields

$$V' \le -\{\theta/2 - 2(e^\theta - 1 - \theta)\}V + \theta \le -\theta V/4 + \theta.$$

Bounded birth rates preserve a finite exponential moment on each bounded interval, justifying differentiation by domination. Hence $V \le M$ is invariant whenever $M \ge 4$.

For a fixed mean α, write $C_{\alpha,M} = \{z : \mu(z) = \alpha,\ V_\theta(z) \le M\}$. This set is compact in d_1. Indeed, the exponential bound gives tightness and uniform integrability of first moments for each marginal, so every sequence has a Wasserstein-convergent subsequence. Means pass to the limit, and Fatou's lemma preserves the moment bound. The same argument proves compactness of

$$K_M = \{z : \mu(z) \in [0,1],\ V_\theta(z) \le M\}.$$

For $\alpha \in [0,1]$, identical mixtures of 0 and 1 with mean α have $V_\theta \le 2\cosh\theta < 4$. Thus all these slices are nonempty for $M \ge 4$.

■ SM8.2 Explicit positive lower bounds

Choose integers ℓ, r with $a_\ell(0) = A > 0$ and $b_r(0) = B > 0$. Variation of constants in (SM8.1) gives

$$\begin{aligned} a_k(t) &\ge a_k(0)e^{-2t} + 2\int_0^t e^{-2(t-s)} a_{k-1}(s) y_{k-1}(s)\,\mathrm{d}s, \\ b_k(t) &\ge b_k(0)e^{-2t} + 2\int_0^t e^{-2(t-s)} b_{k-1}(s) x_k(s)\,\mathrm{d}s. \end{aligned} \tag{SM8.2}$$

The initial atoms therefore retain Ae^{-2t} and Be^{-2t}. If $\ell \le r$, use $y_{j-1} \ge b_r$ to generate successively the left atoms at $\ell+1, \dots, r+1$. Then $x_{r+1} \ge a_{r+1}$ generates b_{r+1}. If $\ell > r$, use $x_j \ge a_\ell$ to generate the right atoms at $r+1, \dots, \ell$. In both cases, with $N = \max(\ell, r+1)$, both atoms at N are positive for every $t > 0$. From both atoms at k, first generate a_{k+1} using b_k, and then generate b_{k+1} using a_{k+1}. This proves positivity at every $k \ge N$ for all $t > 0$.

This induction has quantitative bounds, rather than only a support argument. Suppose two factors in a birth flux have lower bounds $Ct^p e^{-\gamma t}$ and $Dt^q e^{-\eta t}$, with $C, D > 0$ and $p, q \ge 0$. Equation (SM8.2) bounds the generated atom below by

$$\frac{2CD}{p+q+1}\, t^{p+q+1} e^{-(2+\gamma+\eta)t}. \tag{SM8.3}$$

Starting with Ae^{-2t} and Be^{-2t} yields explicit positive constants at every finitely generated level. On $[a,b] \subset (0,\infty)$ such a bound is at least $Ca^p e^{-\gamma b} > 0$. A high atom also bounds each lower finite tail from below. None of these constants is asserted uniform in the seeds A, B, the levels, or the initial laws.

■ SM8.3 A common state and strict cancellation

For two flows, set $d_k = x_k - \widetilde{x}_k$, $e_k = y_k - \widetilde{y}_k$ and use bars for arithmetic midpoints. Bilinear expansion gives exactly

$$\begin{aligned} d_k' &= 2\bar{y}_{k-1}(d_{k-1} - d_k) + 2\bar{a}_{k-1} e_{k-1}, \\ e_k' &= 2\bar{x}_k(e_{k-1} - e_k) + 2\bar{b}_{k-1} d_k. \end{aligned}$$

This is the forward equation of the main article's midpoint chain. Its total exit rates from X_k, Y_k are $2\bar{y}_{k-1}, 2\bar{x}_k \le 2$. Its transition kernel $P_{0,t}(i,j)$ is positive and mass preserving. Bounded rates justify its ordered-jump expansion on the countable state space and the resulting nonexpansion on signed ℓ_1 masses.

Fix $t > 0$ and two finite starting states. Choose $0 < a < b < t$. The previous subsection makes all same-channel forward rates positive on $[a,b]$ and all crossing rates positive at sufficiently high levels. Choose K above both starting levels with $K-1$ in that positive region. An X state can move forward to X_K; a Y state can move forward to Y_{K-1} and then cross to X_K. If one such path has L edges with rate lower bounds $c_1, \dots, c_L > 0$ on $[a,b]$, its ordered-jump integral gives

$$P_{0,t}(i, X_K) \ge e^{-2t}\, \frac{(b-a)^L}{L!} \prod_{j=1}^{L} c_j > 0. \tag{SM8.4}$$

The exponential accounts for no unlisted jumps before, between, or after the prescribed jumps. For $L = 0$, use the no-jump probability e^{-2t}. The rate bounds are supplied by (SM8.3), including the midpoint factor $1/2$ if only one of the two flows is used.

Let $v = (d, e)$ initially. Equal means imply $\sum_i v_i = 2(\mu - \widetilde{\mu}) = 0$. If $v \ne 0$, choose i, j with $v_i > 0$ and $v_j < 0$. Write $r = v^+ P_{0,t}$ and $s = v^- P_{0,t}$. Positivity and mass preservation give

$$\begin{aligned} \|vP_{0,t}\|_1 &= \|v\|_1 - 2\sum_k \min(r_k, s_k) \\ &\le \|v\|_1 - 2\min\{v_i P_{0,t}(i, X_K), (-v_j) P_{0,t}(j, X_K)\} < \|v\|_1. \end{aligned} \tag{SM8.5}$$

This proves strict distance decrease for every positive time and every distinct equal-mean pair with finite first moments. It does not supply a uniform contraction factor smaller than one.

SM8.4 Fixed points and uniform attraction

For $M \geq 4$, the continuous map $T = \mathsf{S}_{-1}\Phi_1$ preserves $C_{\alpha,M}$. It is nonexpansive and strictly shortens distances between distinct points. Here are all steps of the compact fixed-point argument. For $z \in C_{\alpha,M}$ let $a_n = d_1(T^n z, T^{n+1} z) \downarrow r$. Take a convergent subsequence $T^{n_j} z \to w$. Continuity gives both $d_1(w, Tw) = r$ and $d_1(Tw, T^2 w) = r$. Strictness forces $Tw = w$, and hence $r = 0$. Two distinct fixed points contradict strictness, so the fixed point is unique. For any initial z, let $d_1(T^n z, w) \downarrow l$. A convergent subsequence with limit v gives $d_1(v, w) = d_1(Tv, w) = l$. If $v \neq w$, strictness contradicts equality. Thus $l = 0$ and every orbit converges to w.

Take the fixed point z_* in $C_{0,4}$ and set $\rho^u = \mathsf{S}_{\lfloor u \rfloor}\Phi_{\{u\}} z_*$. The identity $\Phi_1 z_* = \mathsf{S}_1 z_*$ and commutation with integer shifts give continuity across integer phases, $\rho^{u+1} = \mathsf{S}_1 \rho^u$, and $\Phi_t \rho^u = \rho^{u+t}$. The moment bound gives $V_\theta(\rho^u) \leq 4$. Each ρ^α is the fixed point in every $C_{\alpha,M}$, $M \geq 4$. Any other finite-first-moment periodic family has a fixed point on the same mean slice; strictness rules it out without assuming its exponential moments.

For $z \in K_M$, put

$$f_n(z) = d_1(T^n z, \rho^{\mu(z)}).$$

The mean map and the family ρ^u are continuous, so each f_n is continuous. Nonexpansion and $T\rho^{\mu(z)} = \rho^{\mu(z)}$ give $f_{n+1} \leq f_n$. The preceding slice argument gives $f_n(z) \to 0$ for every $z \in K_M$. For $\delta > 0$ the relatively open sets $U_n = \{z \in K_M : f_n(z) < \delta\}$ are increasing and cover K_M. Compactness yields a finite subcover. Taking its largest index N gives $K_M = U_N$, and therefore $\sup_{z \in K_M} f_n(z) \leq \delta$ for every $n \geq N$. This proves uniform convergence explicitly. For $t = n + s$, $0 \leq s < 1$, shift invariance and nonexpansion imply

$$d_1(\Phi_t z, \rho^{\mu(z)+t}) = d_1(\Phi_s T^n z, \Phi_s \rho^{\mu(z)}) \leq f_n(z).$$

Integer shifts reduce arbitrary initial means to $[0, 1]$. The argument holds for each fixed M, and gives no rate uniform as M grows. The only uniformity step is compactness of K_M; no uniform minorization is used.

Finally, all orbit atoms are positive. Starting from any phase u, the positivity argument gives a level N above which atoms are positive at every later time. For an arbitrary k, choose an integer $n > 0$ with $k + n \geq N$. Then the atom at $k + n$ in ρ^{u+n} is positive, and the period relation identifies it with the atom at k in ρ^u. Each marginal increases stochastically along the birth flow. For $u \leq v$, its Wasserstein distance is its mean increment. Summing the two increments gives $d_1(\rho^u, \rho^v) = 2(v - u)$, completing the profile theorem.

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